

AX60 Clinical Audiometer — Hardware & Feature GPIO Control Specification

This document is the authoritative hardware and firmware reference for controlling every feature and audio routing mode on the **AX60 Audiometer (STM32F446VET6)**.

1. Master GPIO Pin Mapping Table

Pin Name	Port / Pin	Alternate / Mode	Active Level	Default State	Description / Hardware Destination
AC_Left_EN	PE13	GPIO Output	HIGH	LOW	Air Conduction (AC) Left Earphone Relay (TDH-39 / DD45)
AC_Right_EN	PE14	GPIO Output	HIGH	LOW	Air Conduction (AC) Right Earphone Relay
BC_EN	PE7	GPIO Output	HIGH	LOW	Bone Conduction (BC) Transducer Relay (Radioear B71 / B81)
INSERT_EP_EN	PE12	GPIO Output	HIGH	LOW	Insert Earphone Relay (ER-3A / ER-5A)
FF_EN	PE9	GPIO Output	HIGH	LOW	Free Field (FF) Loudspeaker Output Relay
WN_EN / MH_EN	PE8	GPIO Output	HIGH	LOW	White Noise / Monitor Headphone Enable Relay
INTERNAL_SPK_EN	PA15	GPIO Output	HIGH	LOW	Internal Cabinet Speaker Relay (Examiner Talkback)
OPA_EN	PB2	GPIO Output	HIGH	HIGH	High-Current Analog Output Op-Amp Enable
TALKOVER_EN	PD4	GPIO Output	HIGH=Ext, LOW=Int	LOW	Talkover Mode: Mic Selector (HIGH=External, LOW=Internal)
MIC_AUX_EN	PD5	GPIO Output	HIGH	LOW	Auxiliary 3.5mm Stereo Audio Input Buffer Enable
MIC_EN	PD6	GPIO Output	HIGH	LOW	Patient Talkback Microphone Pre-Amp Enable
PCM_MIC_Control	PA9	GPIO Output	HIGH=Mic/Aux, LOW=DAC	LOW	Analog Multiplexer: Selects Mic/Aux (HIGH) or PCM DAC Tone (LOW)
PCM1808_DATA_EN	PD10	GPIO Output	HIGH	LOW	Audio ADC (PCM1808) Serial Data Output Enable
PGA_MUTE_1	PB13	GPIO Output	LOW=Mute, HIGH=Active	LOW (Muted)	PGA2311 Hardware Output Mute Control
ZCEN	PB14	GPIO Output	HIGH=Zero Cross, LOW=Imm	LOW	PGA2311 Zero-Crossing Gain Step Synchronizer
PGA_CS1	PB12	GPIO Output	Active LOW	HIGH	PGA2311 SPI Chip Select
PGA_SCLK	PB10	GPIO Output	Rising Edge Clock	LOW	PGA2311 SPI Serial Clock
PGA_SDI	PC1	GPIO Output	Data In (MOSI)	LOW	PGA2311 Serial Data Input
PGA_SDO	PC2	GPIO Input	Data Out (MISO)	High-Z	PGA2311 Serial Data Output
PAT_RESPONSE_SW	PB3	EXTI3 Input	Active LOW	HIGH (Pull-up)	Patient Push-Button Response Handswitch
BAT_READ	PA0	ADC1_IN0	Analog	Analog In	Battery Voltage Divider Sensing
POWER_DETECTION	PA1	GPIO Input	HIGH=Connected	In	DC Mains / Wall Adapter Power Detector
RTC_EN (CE)	PB1	GPIO Output	Active HIGH	LOW	DS1302 Real-Time Clock Chip Enable
RTC_CLK (SCLK)	PB6	GPIO Output	Rising Edge	LOW	DS1302 Serial Clock
RTC_IO (DAT)	PB9	GPIO Bidirectional	Data Line	In/Out	DS1302 Bidirectional Data Pin
FRAM_CS	PC13	GPIO Output	Active LOW	HIGH	SPI3 Non-Volatile FRAM Chip Select (Calibration Tables)
SD_CS	PC12	GPIO Output	Active LOW	HIGH	SPI3 MicroSD Card Chip Select (Patient Records)
USART2 TX / RX	PA2 / PA3	AF7 (USART2)	Serial TTL	--	Proculus 7-inch Intelligent Display (115200 8N1)
USART3 TX / RX	PD8 / PC5	AF7 (USART3)	Serial TTL	--	CH340G USB Serial (WebUI / Future Thermal Printer)

2. Feature-by-Feature GPIO Control Matrix

Quick Reference Summary Table

Feature Mode	PCM_MIC_Control (PA9)	TALKOVER_EN (PD4)	MIC_EN (PD6)	MIC_AUX_EN (PD5)	AC_Left (PE13)	AC_Right (PE14)	BC_EN (PE7)	INSERT_EP (PE12)	FF_EN (PE9)	MH_EN (PE8)	INTERNAL_SPK (PA15)	OPA_EN (PB2)
1. Talkover (Ext Mic)	SET (1)	SET (1)	0	0	1	1	0	0	0	0	0	1
2. Talkover (Int Mic)	SET (1)	RESET (0)	0	0	1	1	0	0	0	0	0	1
3. Talkback (Cabinet Spk)	SET (1)	0	SET (1)	0	0	0	0	0	0	0	SET (1)	1
4. Talkback (Headphones)	SET (1)	0	SET (1)	0	0	0	0	0	0	SET (1)	0	1
5. AUX Audio Input	SET (1)	0	0	SET (1)	1	1	0	0	0	0	0	1
6. AC Pure Tone (Left)	RESET (0)	0	0	0	1	1*	0	0	0	0	0	1
7. AC Pure Tone (Right)	RESET (0)	0	0	0	1*	1	0	0	0	0	0	1
8. Bone Conduction (BC)	RESET (0)	0	0	0	0	0	1	1**	0	0	0	1
9. Insert Earphone	RESET (0)	0	0	0	0	0	0	1	0	0	0	1
10. Free Field (Spk)	RESET (0)	0	0	0	0	0	0	0	1	0	0	1
11. Standby / Mute	RESET (0)	0	0	0	0	0	0	0	0	0	0	1

*Note: In AC testing, opposite ear relay is turned ON when contralateral masking noise is active.
**Note: In BC testing, INSERT_EP_EN is turned ON to deliver contralateral narrowband masking noise.

3. Detailed Operation & Code Implementation for Each Feature

Feature 1: Talkover (Examiner → Patient)

- **Goal:** Examiner speaks into the console mic to communicate instructions directly into the patient's headphones.
- **GPIO States:**
 - `PCM_MIC_Control` (`PA9`) → **HIGH (1)** (Bypasses DAC tone generator; routes analog mic pre-amp)
 - `TALKOVER_EN` (`PD4`):
 - **HIGH (1)** = External Gooseneck / Headset Microphone
 - **LOW (0)** = Internal Front-Panel Console Microphone
 - `MIC_EN` (`PD6`) → **LOW (0)** (Patient mic disabled)
 - `MIC_AUX_EN` (`PD5`) → **LOW (0)** (Auxiliary line disabled)
 - `AC_Left_EN` (`PE13`) & `AC_Right_EN` (`PE14`) → **HIGH (1)** (Both patient headphone sides enabled)
 - `BC_EN` (`PE7`), `INSERT_EP_EN` (`PE12`), `FF_EN` (`PE9`) → **LOW (0)**
 - `MH_EN` (`PE8`), `INTERNAL_SPK_EN` (`PA15`) → **LOW (0)** (Examiner speakers muted)
 - `OPA_EN` (`PB2`) → **HIGH (1)** (Output op-amp active)
 - **PGA Volume:** Set to unity gain (`192`) = 0 dB) via `PGA2311_SetVolume(192, 192)` .

```
void Feature_Enable_Talkover(uint8_t use_external_mic)
{
    /* 1. Mute PGA to prevent audio pop */
    PGA2311_Mute();

    /* 2. Select Mic over DAC */
    HAL_GPIO_WritePin(GPIOA, PCM_MIC_Control_Pin, GPIO_PIN_SET);

    /* 3. Choose External vs Internal Examiner Mic */
    HAL_GPIO_WritePin(GPIOD, TALKOVER_EN_Pin, use_external_mic ? GPIO_PIN_SET : GPIO_PIN_RESET);
    HAL_GPIO_WritePin(GPIOD, MIC_EN_Pin | MIC_AUX_EN_Pin, GPIO_PIN_RESET);

    /* 4. Route audio strictly to patient headphones */
    HAL_GPIO_WritePin(GPIOE, AC_Left_EN_Pin | AC_Right_EN_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(GPIOE, BC_EN_Pin | INSERT_EP_EN_Pin | FF_EN_Pin | MH_EN_Pin, GPIO_PIN_RESET);
    HAL_GPIO_WritePin(GPIOA, INTERNAL_SPK_EN_Pin, GPIO_PIN_RESET);

    /* 5. Set unity gain and unmute */
    PGA2311_SetVolume(192, 192);
    PGA2311_Unmute();
}
```

Feature 2: Talkback (Patient → Examiner)

- **Goal:** Allows patient inside a soundproof booth to speak to the examiner. Prevents acoustic feedback howling by isolating patient transducers.

- **GPIO States:**
- `PCM_MIC_Control` (`PA9`) → **HIGH (1)**
- `MIC_EN` (`PD6`) → **HIGH (1)** (Activates patient booth mic amplifier)
- `TALKOVER_EN` (`PD4`) → **LOW (0)**
- `MIC_AUX_EN` (`PD5`) → **LOW (0)**
- **Patient Isolation:** `AC_Left_EN` =0, `AC_Right_EN` =0, `BC_EN` =0, `INSERT_EP_EN` =0, `FF_EN` =0
- **Examiner Destination:**
 - **Cabinet Speaker:** `INTERNAL_SPK_EN` (`PA15`) → **HIGH (1)**, `MH_EN` (`PE8`) → **LOW (0)**
 - **Monitor Headphones:** `MH_EN` (`PE8`) → **HIGH (1)**, `INTERNAL_SPK_EN` (`PA15`) → **LOW (0)**

```
void Feature_Enable_Talkback(uint8_t destination_is_internal_speaker)
{
    PGA2311_Mute();

    /* Select Patient Mic */
    HAL_GPIO_WritePin(GPIOA, PCM_MIC_Control_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(GPIOD, MIC_EN_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(GPIOD, TALKOVER_EN_Pin | MIC_AUX_EN_Pin, GPIO_PIN_RESET);

    /* Mute all patient transducers to prevent feedback loop */
    HAL_GPIO_WritePin(GPIOE, AC_Left_EN_Pin | AC_Right_EN_Pin | BC_EN_Pin | INSERT_EP_EN_Pin | FF_EN_Pin, GPIO_PIN_RESET);

    /* Route to Examiner Monitor */
    if (destination_is_internal_speaker) {
        HAL_GPIO_WritePin(GPIOA, INTERNAL_SPK_EN_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(GPIOE, MH_EN_Pin, GPIO_PIN_RESET);
    } else {
        HAL_GPIO_WritePin(GPIOE, MH_EN_Pin, GPIO_PIN_SET);
        HAL_GPIO_WritePin(GPIOA, INTERNAL_SPK_EN_Pin, GPIO_PIN_RESET);
    }

    PGA2311_SetVolume(192, 192);
    PGA2311_Unmute();
}
```

Feature 3: Auxiliary Audio Input (Speech Audiometry / CD / AUX)

- **Goal:** Feeds pre-recorded speech material from a 3.5mm input jack into the patient's headphones.
- **GPIO States:**
- `PCM_MIC_Control` (`PA9`) → **HIGH (1)**
- `MIC_AUX_EN` (`PD5`) → **HIGH (1)**
- `TALKOVER_EN` (`PD4`) → **LOW (0)**, `MIC_EN` (`PD6`) → **LOW (0)**
- `AC_Left_EN` (`PE13`) & `AC_Right_EN` (`PE14`) → **HIGH (1)**
- `BC_EN` =0, `INSERT_EP_EN` =0, `FF_EN` =0, `MH_EN` =0, `INTERNAL_SPK_EN` =0
- `OPA_EN` (`PB2`) → **HIGH (1)**

Feature 4: Air Conduction Pure Tone & Warble (AC Left / Right)

- **Goal:** Delivers pure tones (125 Hz to 8000 Hz) or Warble tones at -10 dB to 120 dB HL.
- **GPIO States:**
- `PCM_MIC_Control` (`PA9`) → **LOW (0)** (Selects I2S1 PCM5102 DAC output)
- `TALKOVER_EN` =0, `MIC_EN` =0, `MIC_AUX_EN` =0
- `AC_Left_EN` (`PE13`) → **HIGH (1)**
- `AC_Right_EN` (`PE14`) → **HIGH (1)**
- `BC_EN` =0, `INSERT_EP_EN` =0, `FF_EN` =0
- **PGA Attenuator:** Gain calculated via `DbToPGA2311(db_level)` .

Feature 5: Bone Conduction (BC with Contralateral Masking)

- **Goal:** Routes stimulus vibrations to Radioear B71/B81 transducer; routes Narrowband Noise masking to contralateral insert earphone.
- **GPIO States:**
- `PCM_MIC_Control` (`PA9`) → **LOW (0)**
- `BC_EN` (`PE7`) → **HIGH (1)** (Enables Bone Conductor power driver)
- `INSERT_EP_EN` (`PE12`) → **HIGH (1)** (Delivers masking noise into opposite ear canal)
- `AC_Left_EN` =0, `AC_Right_EN` =0, `FF_EN` =0

```
void Feature_Enable_BoneConduction(uint8_t enable_insert_masking)
{
    PGA2311_Mute();

    HAL_GPIO_WritePin(GPIOA, PCM_MIC_Control_Pin, GPIO_PIN_RESET);
    HAL_GPIO_WritePin(GPIOE, AC_Left_EN_Pin | AC_Right_EN_Pin | FF_EN_Pin, GPIO_PIN_RESET);

    /* Turn ON Bone Conductor */
    HAL_GPIO_WritePin(GPIOE, BC_EN_Pin, GPIO_PIN_SET);

    /* Enable Insert Earphone for contralateral masking noise */
    HAL_GPIO_WritePin(GPIOE, INSERT_EP_EN_Pin, enable_insert_masking ? GPIO_PIN_SET : GPIO_PIN_RESET);

    PGA2311_Unmute();
}
```

Feature 6: Free Field Testing (FF)

- **Goal:** Directs audio to external sound room amplifier and calibrated speakers.
- **GPIO States:**
- `PCM_MIC_Control` (`PA9`) → **LOW (0)**
- `FF_EN` (`PE9`) → **HIGH (1)**
- `AC_Left_EN` =0, `AC_Right_EN` =0, `BC_EN` =0, `INSERT_EP_EN` =0

4. PGA2311 Attenuation & Click-Free Switching Discipline

The PGA2311 offers 0.5 dB step resolution (0..255). Step 0 = Mute, Step 1 = -95.5 dB, Step 192 = 0 dB (Unity Gain), Step 255 = +31.5 dB.

Safe Pop/Click Elimination Sequence

Always execute this exact 5-step sequence when changing transducers, modes, or stepping gain:

```
[1. Assert PGA_MUTE_1 = LOW]
|
[2. Settle Delay (2 to 5 ms)]
|
[3. Toggle Relay GPIOs (AC_EN, BC_EN, etc.)]
|
[4. Shift 16-bit Volume to PGA2311 via SPI]
|
[5. Deassert PGA_MUTE_1 = HIGH (Unmute)]
```

5. Patient Response Switch & Power Monitoring

Patient Handswitch (`PB3` / `EXTI3`)

- **Pin:** `PB3` configured as input with `GPIO_PULLUP`.
- **Logic:** Active LOW. Pressing the button pulls `PB3` to GND.
- **Handling:** Configured with `EXTI3_IRQHandler`. Fast software debounce window (> 50 ms). Transmits response packet `0xAA 0x50 0x01 0x55` immediately to WebUI and Proculus display.

Battery & Mains Voltage Detection

- `BAT_READ` (`PA0` / `ADC1_IN0`): Measures battery pack voltage through a resistor divider. Polled periodically, averaged, and passed through a 1% slew rate limiter in `battery.c`.
- `POWER_DETECTION` (`PA1`): Digital input. HIGH indicates external 12V/15V DC charger is plugged in.

6. Communication Architecture

Interface	Baud Rate	Frame Protocol	Primary Use Case
USART2 (<code>PA2</code> / <code>PA3</code>)	115200 8N1	Proculus DGUS / UnicView AD (<code>0x5A 0xA5 ...</code>)	Proculus 7-inch Intelligent Display
USART3 (<code>PD8</code> / <code>PC5</code>)	115200 8N1	Binary/ASCII Protocol (<code>0xAA ... 0x55</code>)	CH340G USB Serial (WebUI control / future Thermal Printer)
USB OTG FS	12 Mbps	USB CDC Virtual COM Port	Direct PC USB connection